

It's not just the students who do not understand issues correctly; sadly, some professors don't, either. Sometimes students get flamed when they try to engage with a FOSS project, which takes a toll too. Open source projects are large and complex, and often not well documented. Code reading skills need to come early, not just code writing. Also, if you're working all alone, then it doesn't quite fit into the spirit of FOSS.

Some students in the US argue, "My dad asks me how I can pay back my student loans by working on open source software." This indicates a misunderstanding about the career paths and opportunities within FOSS.

Likins and Bernius point out to certain flaws within higher education itself—it takes quite some time to start new courses; it can be very hard to teach FOSS (which is barely mentioned in the curriculum); teachers have too much content to wade through; most instructors themselves lack FOSS experience, etc. Sharing code is, sadly, often considered as 'cheating'. So that ends up as a disincentive for working with others, which is quite against the spirit of FOSS.

The timelines in a typical FOSS project do not match with the typical schedules within academic circles. Things change much faster in the FOSS world, which makes it tough for teachers in higher education.

Incidentally, groups like Mozilla are working to enhance students' participation in FOSS projects. For instance, in June 2018, Mozilla's Open Source Student Network (OSSN) announced plans for a pilot programme to support university students working towards their first code contribution (<http://bit.ly/MozillaInitiative>).

Mozilla believes that the most valuable contributions to a project often come from people under the age of 30. FOSS developers need to start young. But students who want to contribute often feel intimidated, feel that they don't have the appropriate skills, or are unable to find the right projects to work on.

To overcome such situations, Likins and Bernius suggest that FOSS can take steps and act "as a community." FOSS needs programmes that train instructors on how to teach open source. It needs to directly engage with college students and mentor them. Above all, it also needs to make its communities more welcoming. Workshops, curriculum material and actual projects could help.

"To mentor, you don't need lead tech skills. All you need is empathy," they remind us. Their advice is to begin editing and writing. They advise newbies to undertake learning activities and evangelise at every opportunity, including at places like a science club. Reporting bugs, documentation and making small code fixes create a positive impact. One need not descend directly into coding.

For those established in FOSS, Likins and Bernius suggest not working just with students, but with professors too, who ensure continuity in the system. More important are professors who are smart enough to work around the system.

Clearly, the attitude against collaboration has to change. One solution could be to get members of the alumni to return to a university and talk about FOSS.

Given this situation in the West (primarily the US), how are things in India? *OSFY* approached Dr Nagarjuna Gadiraju, the current chairman of the Free Software Foundation - India, and someone who has taken a long-term interest in promoting FOSS in education. Dr Nagarjuna works in the Homi Bhabha Centre for Science Education (HBCSE), Tata Institute of Fundamental Research in Mumbai. His major research interests include science education, cognitive science, the history and philosophy of science, and the structure and dynamics of knowledge. As an activist, he focuses on promoting free knowledge and free software.

Dr Nagarjuna believes that the main stumbling blocks in the path to adopting FOSS in Indian academia are the "...use of brand names in the syllabus, and assessments being done only on non-FOSS applications." He points out that the national ICT Policy, which is already out, favours FOSS. So, he argues, state school boards need to follow the guidelines of this policy.

In India, ICT in schools is mostly limited to what students are exposed to in the computer lab, to which children are taken in batches. Except in elite schools, there is really no ICT in classrooms, Dr Nagarjuna argues. He adds: "The lack of FOSS in schools is due to a lack of awareness. Most ICT training programmes focus on proprietary software, which is meant for clerical tasks. Or they use applications that deliver content, using computers."

But there are some notable exceptions too, Dr Nagarjuna points out. The knowledge lab of the HBCSE, with its Connected Learning Initiative (CLIX), is taking FOSS to four Indian states –Telangana, Rajasthan, Chattisgarh and Mizoram. CLIX is a technology-enabled initiative for high school students. Seeded by the Tata Trusts, it is led by Tata Institute of Social Sciences, Mumbai and MIT (Massachusetts Institute of Technology).

"CLIX links pedagogical design and leverages contemporary technology and online capabilities," says Dr Nagarjuna. Resources for students have been created in mathematics, sciences, communicative English and digital literacy, all released as open educational resources (OERs). Teacher professional development is available through professional communities of practice and a post-graduate certificate in Reflective Teaching with ICT.

"What we are doing is to provide a school with a server that has plenty of open educational resources as well as software for activity based education," Dr Nagarjuna explains.

The Decentralised Distributable Disk of Offline Open Educational Resources (DOER) links to plenty of resources and can be checked via its site (<http://doer.metaStudio.org/>). Providing a school server is a starting point; it should be followed up by empowering the teachers, argues Dr Nagarjuna.

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